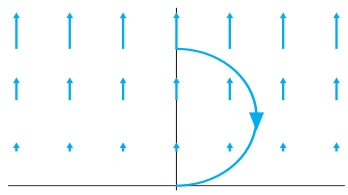
Suggested problems in §18.1: 1-29 odd

Suggested problems in §18.2: 5-29 odd

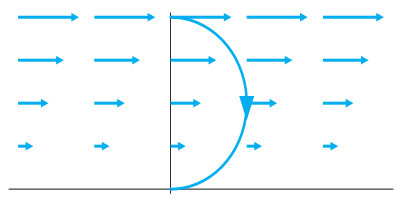
Problem 1: For each of the paths and vector fields below, state whether you expect the line integral along the curve to be positive, negative, or zero.

(a)



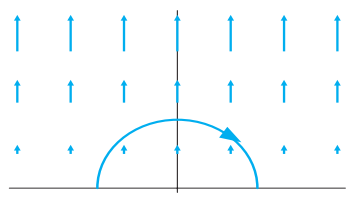
negative

(b)



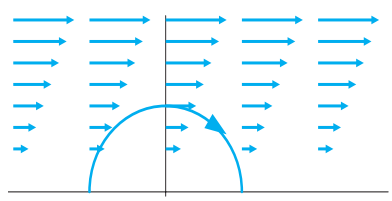
positive

(c)



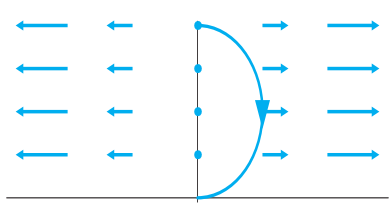
zero

(d)



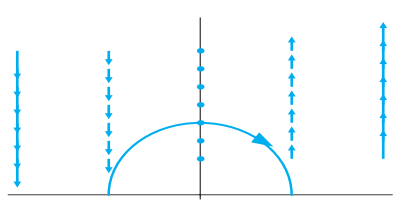
positive

(e)



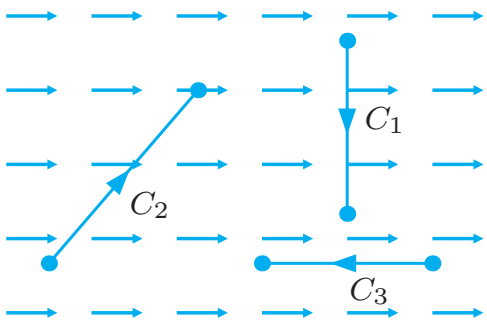
zero

(f)



negative

Problem 2: Consider the vector field $\vec{F}$ shown below together with the paths C_1, C_2, and C_3. Arrange the line integrals over the curves in ascending order.



$$\int_{C_3} \vec{F} \cdot d\vec{r} < \int_{C_1} \vec{F} \cdot d\vec{r} < \int_{C_2} \vec{F} \cdot d\vec{r}.$$

Problem 3: Compute the line integrals.

(a) $\int_C \langle x+y, y\rangle \cdot d\vec{r}$ where C is the quarter unit circle from $(1,0)$ to $(0,1)$.

A parameterization is

$$\vec{r}(t) = \langle \cos(t), \sin(t)\rangle, \quad t \in [0, \pi/2].$$

So

$$\begin{aligned}
\int_C \vec{F} \cdot d\vec{r} &= \int_0^{\pi/2} \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\ dt \\
&= \int_0^{\pi/2} \langle \cos(t) + \sin(t), \sin(t)\rangle \cdot \langle -\sin(t), \cos(t)\rangle\ dt \\
&= -\int_0^{\pi/2} \sin^2(t)\ dt \\
&= -\int_0^{\pi/2} \left(\frac{1}{2} - \frac{1}{2}\cos(2t)\right)\ dt \\
&= -\frac{\pi}{4}.
\end{aligned}$$

(b) $\int_C \langle 3, y+5\rangle \cdot d\vec{r}$ where C is the line from $(0,0)$ to $(0,3)$.

A parameterization is

$$\vec{r}(t) = \langle 0, t\rangle, \quad t \in [0,3].$$

So

$$\begin{aligned}
\int_C \langle 3, y+5\rangle \cdot d\vec{r} &= \int_0^3 \langle 3, t+5\rangle \cdot \langle 0, 1\rangle\ dt \\
&= \int_0^3 (t+5)\ dt \\
&= \frac{39}{2}.
\end{aligned}$$

(c) $\int_C \langle x^2+y, y^3, 0\rangle \cdot d\vec{r}$ where C consists of the two line segments from $(4,0,0)$ to $(4,3,0)$ to $(0,3,0)$.

We need to parameterize the two lines, which we call C_1 and C_2. They are

$$\vec{r}_1(t) = \langle 4, 3t, 0\rangle, \quad t \in [0,1]$$

and

$$\vec{r}_2(t) = \langle 4-4t, 3, 0\rangle, \quad t \in [0,1].$$

So

$$\begin{aligned}
\int_C \langle x^2+y, y^3, 0\rangle \cdot d\vec{r} &= \int_{C_1} \langle x^2+y, y^3, 0\rangle \cdot d\vec{r} + \int_{C_2} \langle x^2+y, y^3, 0\rangle \cdot d\vec{r} \\
&= \int_0^1 \langle 16+3t, 27t^3, 0\rangle \cdot \langle 0,3,0\rangle \; dt + \int_0^1 \langle (4-4t)^2+3, 27, 0\rangle \cdot \langle -4,0,0\rangle \; dt \\
&= 81\int_0^1 t^3 \; dt - 4\int_0^1 ((4-4t)^2+3) \; dt \\
&= \frac{81}{4} - \frac{100}{3} \\
&= -\frac{157}{12}.
\end{aligned}$$

(d) $\int_C (\sin(x)\vec{i} + \cos(x^2)\vec{j}) \cdot d\vec{r}$ where C is the parabola $y = x^2$ from $(0,0)$ to $(2,4)$.

A parameterization is

$$\vec{r}(t) = \langle t, t^2\rangle, \quad t \in [0,2].$$

So

$$\begin{aligned}
\int_C (\sin(x)\vec{i} + \cos(x^2)\vec{j}) \cdot d\vec{r} &= \int_0^2 (\sin(t)\vec{i} + \cos(t^2)\vec{j}) \cdot (\vec{i} + 2t\vec{j}) \; dt \\
&= \int_0^2 (\sin(t) + 2t\cos(t^2)) \; dt \\
&= -\cos(2) + \sin(4) + 1.
\end{aligned}$$

Problem 4: Compute the line integrals.

(a) $\int_C 3y \; dx + 4x \; dy$ where C is the line from $(1,3)$ to $(5,9)$.

A parameterization is

$$\vec{r}(t) = \langle 1+4t, 3+6t\rangle, \quad t \in [0,1].$$

So

$$\begin{aligned}
\int_C 3y \; dx + 4x \; dy &= \int_0^1 (12(3+6t) + 24(1+4t)) \; dt \\
&= 144.
\end{aligned}$$

(b) $\int_C x\ dx + z\ dy - y\ dz$ where C is the circle of radius 3 in the yz-plane centered at the origin, oriented counterclockwise when viewed from the positive x-axis.

A parameterization is

$$\vec{r}(t) = \langle 0, 3\cos(t), 3\sin(t)\rangle, \quad t \in [0, 2\pi].$$

So

$$\begin{aligned}\int_C x\ dx + z\ dy - y\ dz &= \int_0^{2\pi} (-9\sin^2(t) - 9\cos^2(t))\ dt \\ &= -9\int_0^{2\pi} dt \\ &= -18\pi.\end{aligned}$$